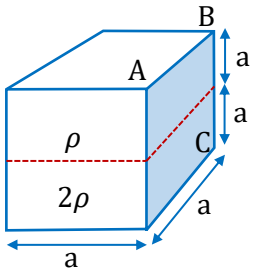
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Pressure and Pascal's law

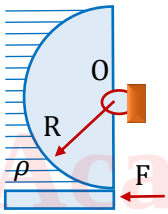
1. A cuboid ($a \times a \times 2a$) is filled with two immiscible liquids of density 2ρ and ρ as shown in the figure. Neglecting atmospheric pressure, ratio of force on base and side wall of the cuboid is:



(A) 2 : 3 (B) 1 : 3 (C) 5 : 6 (D) 6 : 5

PFM099

2. A light semi cylindrical gate of radius R is pivoted at its mid-point O, of the diameter as shown in the figure holding liquid of density ρ. The force F required to prevent the rotation of the gate is equal to



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(A) $2\pi R^3\rho g$ (B) $2\rho g R^3 l$ (C) $\frac{2R^2 l\rho g}{3}$ (D) None of these

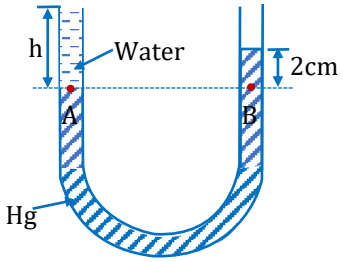
PFM102

3. Some liquid is filled in a cylindrical vessel of radius R. Let F_1 be the force applied by the liquid on the bottom of the cylinder. Now the same liquid is poured into a vessel of uniform square cross-section of side R. Let F_2 be the force applied by the liquid on the bottom of this new vessel. (Neglect atmosphere pressure) Then:

(A) $F_1 = \pi F_2$ (B) $F_1 = \frac{F_2}{\pi}$ (C) $F_1 = \sqrt{\pi}\ F_2$ (D) $F_1 = F_2$

PFM168

4. The liquids shown in figure in the two arms are mercury (density = 13.6 gm/cm^3) and water. If the difference of heights of the mercury column is 2 cm, find the height h of the water column:-



(A) 10 cm (B) 2 cm (C) 18 cm (D) 27.2 cm

PFM169

For More Join: @IITJEE_Advanced

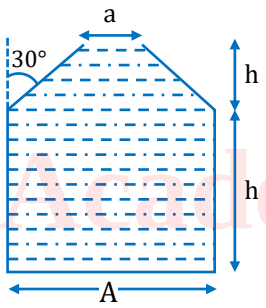
5. Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density ρ. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other is x_2. When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is :

(A) $g\rho S\,(x_2 + x_1)^2$ (B) $\frac{3}{4}g\rho S(x_2 - x_1)^2$

(C) $\frac{1}{4}g\rho S(x_2 - x_1)^2$ (D) $g\rho S(x_2^2 + x_1^2)$

PFM171

6. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section A and height h. The upper part is a conical vessel of height h with base area 'A' and top area 'a' and the walls of the vessel are inclined at an angle 30° with the vertical. A liquid of density ρ fills both the sections upto a height $2h$. Neglecting atmospheric pressure.



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(A) The force F exerted by the liquid on the base of the vessel is $2h\rho g\,\frac{(A+a)}{2}$.

(B) The pressure P at the base of the vessel is $2h\rho g\,\frac{A}{a}$.

(C) The weight of the liquid W is greater than the force exerted by the liquid on the base.

(D) The walls of the vessel exert a downward force $(F - W)$ on the liquid.

PFM172

7. A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 *cm* and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in *cm*, will be:

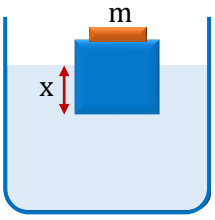
(A) 1.2 (B) 0.1 (C) 2.0 (D) 0.4

PFM108

For More Join: @IITJEE_Advanced

Buoyancy

8. A cubical block ($a \times a \times a$), with a coin of mass 'm' over it is floating over a liquid of density ρ. In this case x_1 depth of the block is immersed. Now the coin is removed and it is found that x_2 depth is immersed in liquid. Value of $(x_1 - x_2)$ is :



(A) $\frac{m}{\rho a^2}$ (B) $\frac{\rho a^4}{m}$ (C) $\frac{m}{2\rho a^2}$ (D) Data insufficient

PFM106

9. An object of density d_o, kept deep inside water of density d_w and released. During the time it moves a vertical distance h with in the water :-

(A) The gravitational potential energy of the water in the vessel increases if $d_0 < d_w$

(B) The gravitational potential energy of the water in the vessel decreases if $d_0 < d_w$

(C) The gravitational potential energy of the object increases if $d_0 > d_w$

(D) The gravitational potential energy of the object decreases if $d_0 < d_w$

PFM107

10. Mercury is poured into a uniform vertical U-tube and water is poured in it. The level of water is the same in both arms. A piece of wood is dropped into one arm and some water, equal in weight to the piece of wood, is added to the other. Then, consider the statements:

(I) The level of mercury will be same in both the arms.

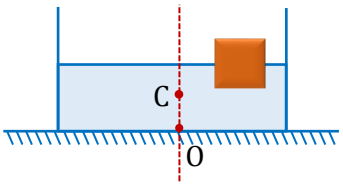
(II) The level of water will be same in both the arms.

(A) Both I and II are true (B) I is true but II is false

(C) I is false but II is true (D) Both I and II are false.

PFM109

11. There is water in container with center of mass at C. Now a small wooden piece is placed towards right as shown in the figure. After putting the wooden piece.



(A) Pressure at base remains same and centre of mass of water and wooden piece will be right of line OC.

(B) Pressure at base remains same and centre of mass of water and wooden piece will be on line OC.

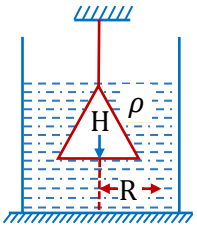
(C) Pressure at base changes and centre of mass of water and wooden piece will be right of line OC.

(D) Pressure at base changes and centre of mass of water and wooden piece will be on line OC.

PFM110

For More Join: @IITJEE_Advanced

12. A cone of radius R and height H is hanging inside a liquid of density ρ by means of a string as shown in this figure. The force, due to the liquid acting on the slant surface of the cone is (Neglect atmosphere pressure)



(A) $\rho\pi gHR^2$ (B) $\pi\rho HR^2$ (C) $\pi\rho gHR^2$ (D) $\frac{2}{3}\pi\rho gHR^2$

PFM114

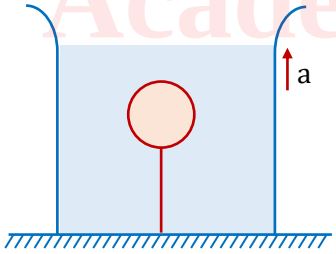
13. A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, form a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is : (Here k is spring constant)

(A) $\frac{Mg}{k}$ (B) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$ (C) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$ (D) $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

PFM116

14. The tension in a string holding a solid block below the surface of a liquid (of density greater than that of solid) as shown in figure is T_0 when the system is at rest. What will be the tension in the string if the system has an upward acceleration a?

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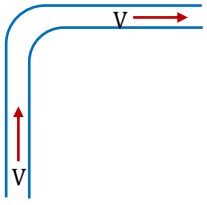


(A) $T_0\left(1+\frac{a}{g}\right)$ (B) $T_0\left(1-\frac{a}{g}\right)$ (C) $2T_0\left(1-\frac{a}{g}\right)$ (D) $3T_0\left(1+\frac{2a}{g}\right)$

PFM117

Fluid dynamics

15. A fire hydrant delivers water of density ρ at a volume rate L. The water travels vertically upward through the hydrant and then does 90° turn to emerge horizontally at speed V. The pipe and nozzle have uniform cross-section throughout. The force exerted by the water on the corner of the hydrant is:-



(A) ρVL (B) zero (C) $2\rho VL$ (D) $\sqrt{2}\rho VL$

PFM122

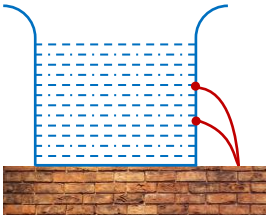
For More Join: @IITJEE_Advanced

16. Water is flowing steadily through a horizontal tube of non-uniform cross-section. If the pressure of water is $4 \times 10^4\ N/m^2$ at a point where cross-section is $0.02\ m^2$ and velocity of flow is $2\ m/s$, what is pressure at a point where cross-section reduces to $0.01\ m^2$.

(A) $1.4 \times 10^4\ N/m^2$ (B) $3.4 \times 10^4\ N/m^2$ (C) $2.4 \times 10^{-4}\ N/m^2$ (D) none of these

PFM123

17. In a cylindrical vessel containing liquid of density ρ, there are two holes in the side walls at heights of h_1 and h_2 respectively such that the range of efflux at the bottom of the vessel is same. Find the height of a hole, for which the range of efflux would be maximum.

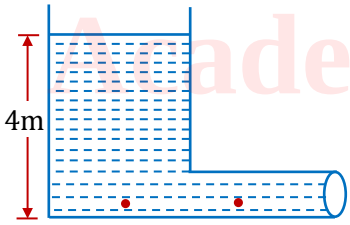


(A) $\left(\frac{h_1}{2}\right)$ (B) $\left(\frac{h_2}{2}\right)$ (C) $\left(\frac{h_2+h_1}{2}\right)$ (D) $(h_2 + h_1)$

PFM124

18. A vent tank of large cross-sectional area has a horizontal pipe $0.12\ m$ in diameter at the bottom. This holds a liquid whose density is $1500\ kg/m^3$ to a height of $4.0\ m$. Assume the liquid is an ideal fluid in laminar flow. In figure, the velocity with which fluid flows out is :-

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(A) $2\sqrt{5}\ m/s$ (B) $\sqrt{5}\ m/s$ (C) $4\sqrt{5}\ m/s$ (D) $\sqrt{10}\ m/s$

PFM127

19. A large tank is filled with water to a height H. A small hole is made at the base of the tank. It takes T_1 time to decrease the height of water to $\frac{H}{\eta}$, ($\eta > 1$) and it takes T_2 time to take out the rest of water.

If $T_1 = T_2$, then the value of η is :

(A) 2 (B) 3 (C) 4 (D) 5

PFM132

Viscosity

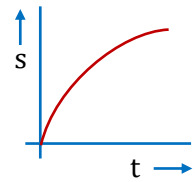
20. There is a $1mm$ thick layer of glycerine between a flat plate of area $100\ cm^2$ and a big fixed plate. If the coefficient of viscosity of glycerine is $1.0\ kg/m\text{-}s$ then how much force is required to move the plate with a velocity of $7\ cm/s$?

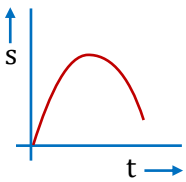
(A) $3.5\ N$ (B) $0.7\ N$ (C) $1.4\ N$ (D) None

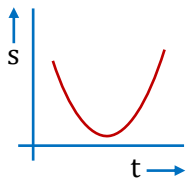
PFM138

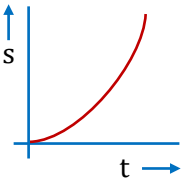
For More Join: @IITJEE_Advanced

21. The displacement of a ball falling from rest in a viscous medium is platted against time. Choose a possible option.

(A) 

(B) 

(C) 

(D) 

PFM139

22. An air bubble of radius 1*cm* is found to rise in a cylindrical vessel of large radius at a steady rate of 0.2 *cm* per second. If the density of the liquid is 1470 $kg\ m^{-3}$, then coefficient of viscosity of liquid is approximately equal to

(A) 163 poise (B) 163 centi-poise (C) 140 poise (D) 140 centi-poise

PFM140

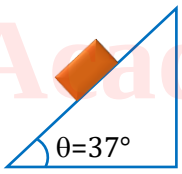
23. A solid metallic sphere of radius r is allowed to fall freely through air. If the frictional resistance due to air is proportional to the cross-sectional area and to the square of the velocity, then the terminal velocity of the sphere is proportional to which of the following?

(A) r^2 (B) r (C) $r^{3/2}$ (D) $r^{1/2}$

PFM145

24. A cubical block of side 'a' and density 'ρ' slides over a fixed inclined plane with constant velocity 'v'. There is a thin film of viscous fluid of thickness 't' between the plane and the block. Then the coefficient of viscosity of the thin film will be:

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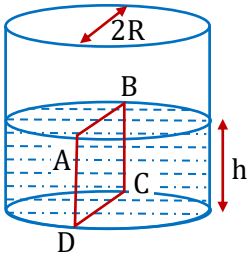


(A) $\frac{3\rho agt}{5v}$ (B) $\frac{4\rho agt}{5v}$ (C) $\frac{\rho agt}{v}$ (D) None of these

PFM147

Surface Tension

25. Water is filled up to a height h in a beaker of radius R as shown in the figure. The density of water is ρ, the surface tension of water is T and the atmospheric pressure is P_0. Consider a vertical section $ABCD$ of the water column through a diameter of the beaker. The force on water on one side of this section by water on the other side of this section has magnitude:



(A) $|2P_0Rh + \pi R^2\rho gh - 2RT|$ (B) $|2P_0Rh + R\rho gh^2 - 2RT|$

(C) $|P_0\pi R^2 + R\rho gh^2 - 2RT|$ (D) $|P_0\pi R^2 + R\rho gh^2 + 2RT|$

PFM149

For More Join: @IITJEE_Advanced

26. Assume that a drop of liquid evaporates by decrease in its surface energy, so that its temperature remains unchanged. What should be the minimum radius of the drop for this to be possible? The surface tension is T, density of liquid is ρ and L is its latent heat of vaporization.

(A) $\frac{\rho L}{T}$ (B) $\sqrt{\frac{T}{\rho L}}$ (C) $\frac{T}{\rho L}$ (D) $\frac{2T}{\rho L}$

PFM153

27. A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is

(Take note that the surface tension applies an upward force on the droplet)

(A) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$ (B) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$ (C) $r = \sqrt{\frac{T}{(d-\rho)g}}$ (D) $r = \sqrt{\frac{T}{(d+\rho)g}}$

PFM154

28. A rigid ring A and a rigid thin disk B both made of same material, when gently placed on water, just manage to float due to surface tension as shown in the figure. Both the ring and the disk have same radius. What can you conclude about their masses?



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(A) Both have the same mass.

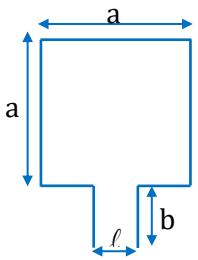
(B) Mass of the ring is half of that of the disk.

(C) Mass of the ring is double to that of the disk.

(D) More information is needed to decide.

PFM155

29. A light open rigid wire frame floats on the surface of water as shown in figure. What force will act on the frame, immediately after some soap solution is dropped inside it? α_1 and α_2 are the surface tensions of water and soap respectively ($\alpha_1 > \alpha_2$).



(A) Zero (B) $(\alpha_1 - \alpha_2)\ell$

(C) $(\alpha_1 + \alpha_2)(4a + 2b + \ell)$ (D) $(\alpha_1 - \alpha_2)(4a + 2b - \ell)$

PFM156

For More Join: @IITJEE_Advanced

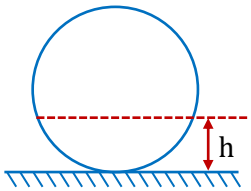
30. When an air bubble rises from the bottom of a deep lake to a point just below the water surface, the pressure of air inside the bubble :-

(A) is greater than the pressure outside it (B) is less than the pressure outside it

(C) increases as the bubble moves up (D) remains same as the bubble moves up

PFM160

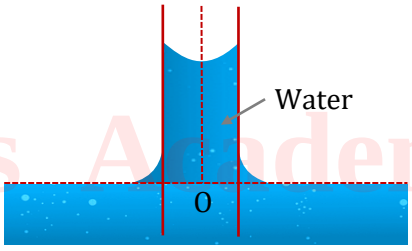
31. A liquid is filled in a spherical container of radius R till a height h. At this position the liquid surface at the edges is also horizontal. The contact angle is :-

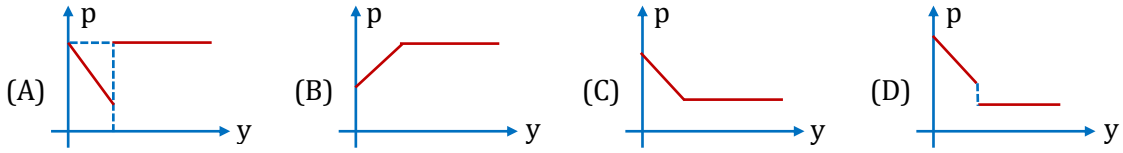


(A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$ (C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$

PFM161

32. Water rises in a capillary as shown. The correct graph of pressure (p) vs height (y) above point O along the axis of capillary is :-





PFM164

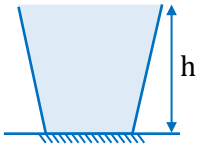
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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. $M\,gm$ of a liquid of density ρ is filled in a light beaker and kept on a horizontal table as shown in the figure. The height of the liquid in the beaker is h. The beaker is wider on top than at its base and the cross-sectional area of the base is A. Neglect the effect of atmospheric pressure. Now, choose the CORRECT statement(s) from the following.



(A) The pressure of liquid at the bottom surface is ρgh.

(B) The normal reaction exerted by the table on the beaker is ρghA.

(C) The pressure of the liquid at the bottom surface is $\frac{Mg}{A}$.

(D) The normal reaction exerted by the table on the beaker is Mg. **PFM100**

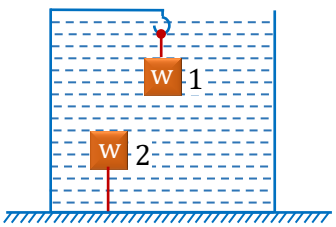
2. A beaker is filled in with water is accelerated by $a\ m/s^2$ in + x direction. The surface of water shall make an angle.

(A) $\tan^{-1}(a/g)$ backwards (B) $\tan^{-1}(a/g)$ forwards

(C) $\cot^{-1}(g/a)$ backwards (D) $\cot^{-1}(g/a)$ forwards

PFM101

3. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.



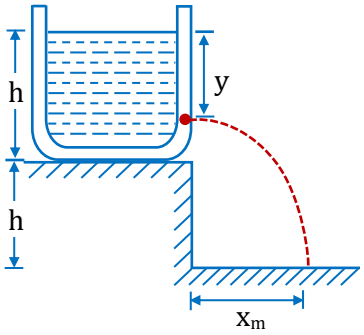
(A) Iron block gets deflected towards right. (B) Wooden block gets deflected towards right.

(C) Iron block gets deflected towards left. (D) Wooden block gets deflected towards left.

PFM119

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For More Join: @IITJEE_Advanced

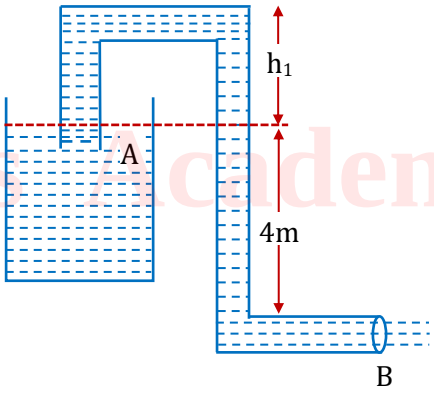
4. A tank is filled upto a height h with a liquid and is placed on a platform of height h from the ground. To get maximum range x_m a small hole is punched at a distance of y from the free surface of the liquid. Then



(A) $x_m = 2h$ (B) $x_m = 1.5\ h$ (C) $y = h$ (D) $y = 0.75\ h$

PFM129

5. A tube of small uniform cross section is used to siphon the water from the vessel. Then choose correct alternative(s) : ($\rho_{water} = 10^3\ kg/m^3, g = 10\ m/s^2, P_{atm} = 10^5\ Pa$)



Rankers Academy JEE

(A) Water will come out from section B with a velocity of $\sqrt{80}\ m/s$

(B) Water will come out from section B with a velocity of $10\sqrt{2}\ m/s$

(C) The greatest value of h_1 for which the siphon will work is $10\ m$

(D) The greatest value of h_1 for which the siphon will work is $6\ m$

PFM131

6. n drops of a liquid, each with surface energy E, joining to form a single drop

(A) some energy will be released in the process

(B) some energy will be absorbed in the process

(C) the energy released or absorbed will be $E(n - n^{2/3})$

(D) the energy released or absorbed will be $nE(n^{2/3} - 1)$

PFM158

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 7 TO 9

When liquid medicine of density ρ is to be put in the eye, it is done with the help of a dropper. As the bulb on the top of the dropper is pressed, a drop forms at the opening of the dropper. We wish to estimate the size of the drop. We first assume that the drop formed at the opening is spherical because that requires a minimum increase in its surface energy. To determine the size, we calculate the net vertical force due to the surface tension T when the radius of the drop is R. When this force becomes smaller than the weight of the drop, the drop gets detached from the dropper.

7. If the radius of the opening of the dropper is r, the vertical force due to the surface tension on the drop of radius R (assuming $r << R$) is

(A) $2\pi rT$ (B) $2\pi RT$ (C) $\frac{2\pi r^2 T}{R}$ (D) $\frac{2\pi R^2 T}{r}$

PFM150

8. If $r = 5 \times 10^{-4}\ m$, $\rho = 10^3\ kgm^{-3}$, $g = 10\ ms^{-2}$, $T = 0.11\ Nm^{-1}$, the radius of the drop when it detaches from the dropper is approximately.

(A) $1.4 \times 10^{-3}\ m$ (B) $3.3 \times 10^{-3}\ m$ (C) $2.0 \times 10^{-3}\ m$ (D) $4.1 \times 10^{-3}\ m$

PFM151

9. After the drop detaches, its surface energy is

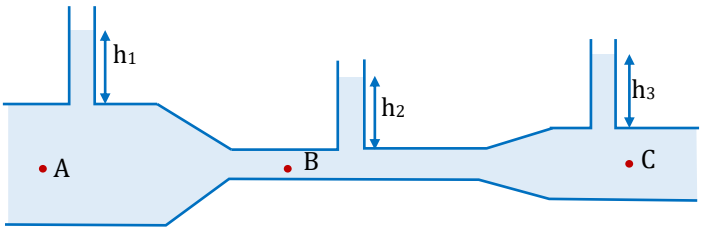
(A) $1.4 \times 10^{-6}\ J$ (B) $2.7 \times 10^{-6}\ J$ (C) $5.4 \times 10^{-6}\ J$ (D) $8.1 \times 10^{-6}\ J$

PFM152

MATRIX MATCH TYPE QUESTION

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10. An ideal liquid is flowing in a tube as shown in figure. Area of cross-section at points A, B and C are A_1, A_2 and A_3 respectively ($A_1 > A_3 > A_2$). v_1, v_2,>v_3 are the velocities at the points A, B and C respectively.



	Column-I		Column-II
(A)	h_1 is	(P)	Less than h_3
(B)	h_2 is	(Q)	More than h_3
(C)	v_1 is	(R)	Less than v_3
(D)	v_2 is	(S)	More than v_3
		(T)	The maximum value amongst the three velocities

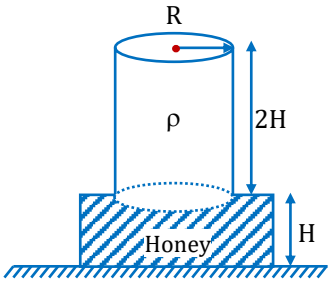
(A) (A-P), (B-Q), (C-S,T), (D-R) (B) (A-R), (B-Q), (C-P), (D-T,S)
(C) (A-Q), (B-P), (C-R), (D-S,T) (D) (A-S,T), (B-Q), (C-P), (D-R)

PFM180

For More Join: @IITJEE_Advanced

MISCELLANEOUS QUESTION

11. A bottle is kept on the ground as shown in the figure. The bottle can be modelled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius if $R\sqrt{2}$ and is filled with honey of density 2ρ. The upper zone of the bottle is filled with the water of density ρ and has a cross-sectional radius R. The height of the lower zone is H while that of the upper zone is $2H$. If now the honey and the water parts are mixed together to form a homogeneous solution: (Assume that total volume does not change)



(A) The pressure inside the bottle at the base will remain unaltered.
(B) The normal reaction on the bottle from the ground will remain unaltered
(C) The pressure inside the bottle at the base will increase by an amount $\left(\frac{1}{2}\right)\rho gH$
(D) The pressure inside the bottle at the base will decrease by an amount $\left(\frac{1}{4}\right)\rho gH$

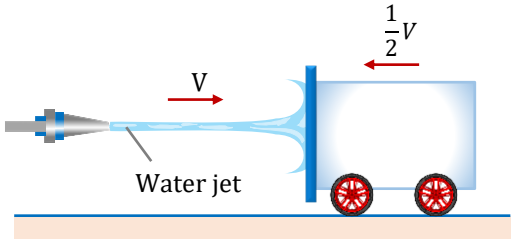
PFM173

12. A solid sphere of radius R and density ρ is attached to one end of a massless spring of force constant k. The other end of the spring is connected to another solid sphere of radius R and density 3ρ. The complete arrangement is placed in a liquid of density 2ρ and is allowed to reach equilibrium. The correct statement(s) is (are):

(A) The net elongation of the spring is $\frac{4\pi R^3\rho g}{3k}$
(B) The net elongation of the spring is $\frac{8\pi R^3\rho g}{3k}$
(C) The light sphere is partially submerged.
(D) The light sphere is completely submerged.

PFM174

13. A horizontal water jet of constant velocity V impinges normally on a vertical flat plate and splashes off the sides in the vertical plane. The plate is moving toward the oncoming water jet with velocity $\frac{1}{2}V$. If a force F is required to maintain the plate stationary, how much force is required to move the plate toward the water jet?



(A) F (B) 1.25 F (C) 2 F (D) 2.25 F

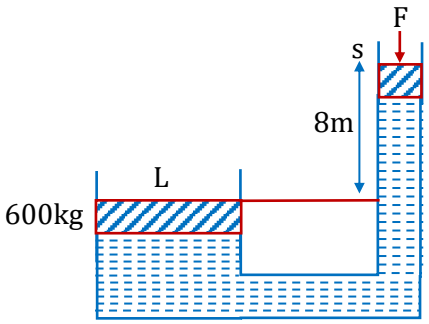
PFM135

For More Join: @IITJEE_Advanced

EXERCISE (S)

Pressure and Pascal's law

1. For the system shown in the figure, the cylinder on the left at L has a mass of 600 kg and a cross-sectional area of 800 cm^2. The piston on the right, at S, has cross sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil. ($\rho = 0.75\ gm/cm^3$)
Find the force F required to hold the system in equilibrium.



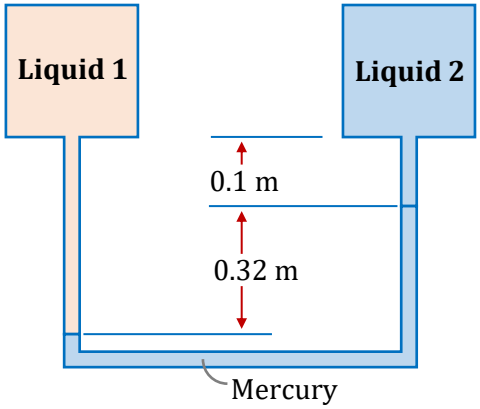
PFM088

2. The volume of an air bubble is doubled as it rises from the bottom of a lake to its surface. If the atmospheric pressure is H m of mercury and the density of mercury is n times that of lake water. Find the depth of the lake.

PFM089

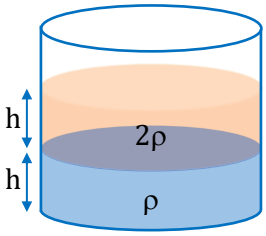
3. A liquid filled in two vessels which are connected by narrow tube as shown in figure. If the mercury level difference between the two hands is 32 cm. Then find pressure difference between the vessels. The density of liquid and mercury is 720 kg/m^3 and 13600 kg/m^3 respectively (g = 9.8).

Bansal's Academy JEE



PFM091

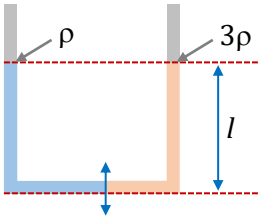
4. As shown, a container is filled with two incompressible liquids with densities ρ and 2ρ. Find the pressure at the bottom of the beaker.



PFM094

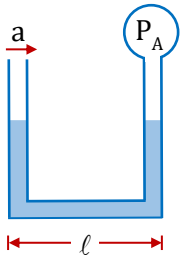
For More Join: @IITJEE_Advanced

5. As shown, the U-shaped tube was filled to height l with two different immiscible liquids of density ρ and 3ρ separated by a valve. If the valve is open, find the new height of the liquid in both columns.



PFM095

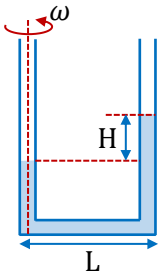
6. A liquid of density ρ is filled in a U-shaped tube, whose one end is open and at the other end a bulb is fitted whose pressure is P_A. Now this tube is moved horizontally with acceleration 'a' as shown in the figure. During the exercise, it was observed that the fluids in the two lines were in the same balance. If the atmospheric pressure is P_0, find the value of P_A.



PFM098

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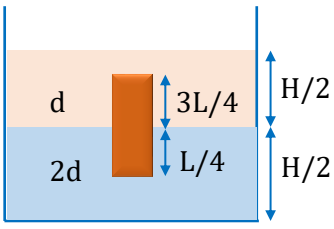
7. A U tube is rotated about one of it's limbs with an angular velocity ω. Find the difference in height H of the liquid (density ρ) level, where diameter of the tube $d << L$.



PFM105

Buoyancy

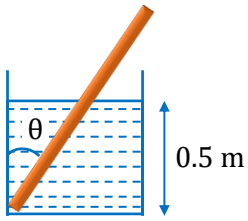
8. A container of a large uniform cross-sectional area A resting on a horizontal surface holds two immiscible, non-viscous and incompressible liquids of densities d and $2d$ each of height $\frac{H}{2}$ as shown. The lower density liquid is open to atmosphere. A homogeneous solid cylinder of length $L\left(<\frac{H}{2}\right)$, cross-sectional area $\frac{A}{5}$ is immersed such that it floats with its axis vertical to the liquid -liquid interface with length $\frac{L}{4}$ in denser liquid. Find the density of the solid cylinder.



PFM111

For More Join: @IITJEE_Advanced

9. A wooden plank of length 1 m and uniform cross-section is hinged at one end to the bottom of a tank as shown. The tank is filled with water upto a height of 0.5 m. The specific gravity of the plank is 0.5. Find the angle θ made by the plank in equilibrium position



PFM113

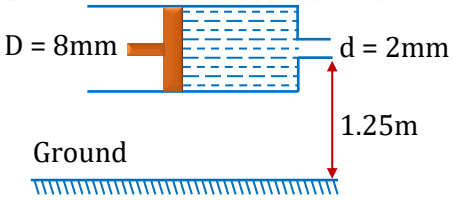
10. A cylindrical wooden float whose base area is 4 m^2 and height 1 m drifts on the water surface in vertical position. Density of wood is 500 kg/m^3 and that of water is 1000 kg/m^3. What minimum work (in kJ) must be performed to take the float out of the water?

PFM118

Fluid dynamics

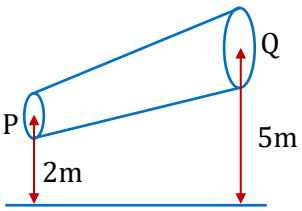
11. Consider a horizontally oriented syringe containing water located at a height of 1.25 m above the ground. The diameter of the plunger is 8 mm and the diameter of the nozzle is 2 mm. The plunger is pushed with a constant speed of 0.25 m/s. Find the horizontal range (in m) of water stream on the ground. Take $g = 10\ m/s^2$.

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PFM125

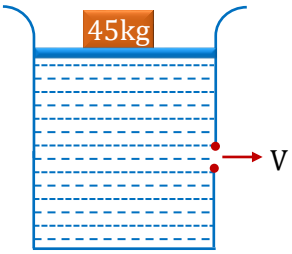
12. A non-viscous liquid of constant density 1000 kg/m^3 flows in a streamline motion along a tube of variable cross section. The tube is kept inclined in the vertical plane as shown in the figure. The area of cross section of the tube at two points P and Q at heights of 2 meters and 5 meters are respectively $4 \times 10^{-3} m^2$ and $8 \times 10^{-3}\ m^3$. The velocity of the liquid at point P is 1 m/s. Find the work done per unit volume by the pressure and the gravity forces as the fluid flows from point P to Q.



PFM126

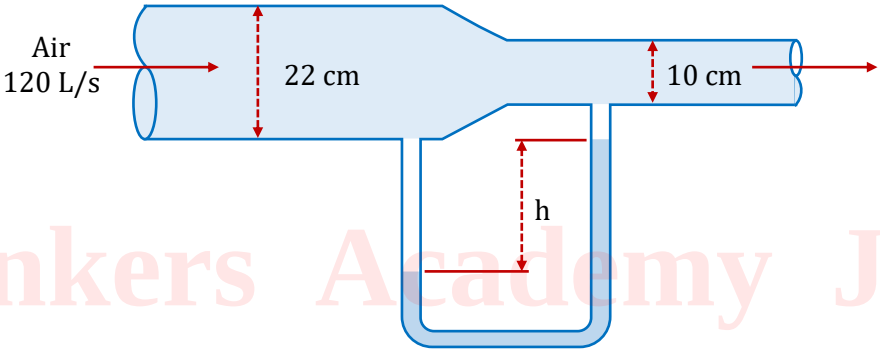
For More Join: @IITJEE_Advanced

13. A large cylindrical tank of cross-sectional area $1m^2$ is filled with water. It has a small hole at a height of $1m$ from the bottom. A movable piston of mass 5 kg is fitted on the top of the tank such that it can slide in the tank freely. A load of 45 kg is applied on the top of water by piston, as shown in figure. Find the value of v (in m/s) when piston is $7m$ above the bottom ($g = 10\ m/s^2$)



PFM130

14. Air flows through a pipe at a rate of 120 L/s. The pipe consists of two sections of diameters 22 cm and 10 cm with a smooth reducing section that connects them. The pressure difference between the two pipe sections is measured by a water manometer. Neglecting frictional effects, determine the differential height (in cm) of water between the two pipe sections. Take the air density to be 1.20 kg/m^3.



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PFM134

15. A large open top container of negligible mass and uniform cross–sectional area A has a small hole of cross–sectional area $A/100$ in its side wall near the bottom. The container is kept on a smooth horizontal floor and contains a liquid of density ρ and mass m_0. Assuming that the liquid starts flowing out horizontal through the hole at $t = 0$, calculate:

(i) the acceleration of the container and

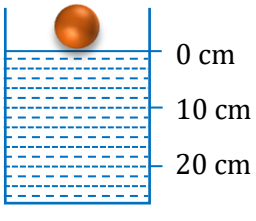
(ii) its velocity when 75% of the liquid has drained out.

PFM137

Viscosity

16. A spherical ball of density ρ and radius $0.003m$ is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 $Ns.m^{-2}$ and its density $\rho_L = \frac{\rho}{2}$ = 1260 $kg.m^{-3}$. Assume the ball reaches a terminal speed by the 10 cm mark. Find the time taken by the ball to traverse the distance between the 10 cm and 20 cm mark.

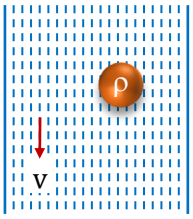
(g = acceleration due to gravity = 10 ms^{-2})



PFM142

For More Join: @IITJEE_Advanced

17. A liquid of density $\rho = 1000\ kg/m^3$ and coefficient of viscosity $h = 0.1\ Ns/m^2$ is flowing down in vertical pipe of large cross section. A small ball of density $\rho_0 = 100\ kg/m^3$ and $r = 5\ cm$ will be at rest in flowing liquid, if velocity of flowing liquid is $10\ k\ m/s$. Then find the value of k.



PFM143

18. A small sphere falls from rest in a viscous liquid. Due to friction, heat is produced. Find the relation between the rate of production of heat and the radius of the sphere at terminal velocity.

PFM144

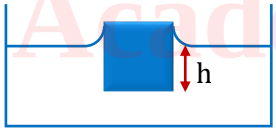
Surface Tension

19. Two arms of a U-tube have unequal diameters $d_1 = 1.0\ mm$ and $d_2 = 1.0\ cm$. If water (surface tension $7 \times 10^{-2}\ N/m$) is poured into the tube held in the vertical position, find the difference of level of water in the U-tube. Assume the angle of contact to be zero.

PFM166

20. A cube with a mass 'm' completely wettable by water floats on the surface of water. Each side of the cube is 'a'. What is the distance h between the lower face of cube and the surface of the water if surface tension is 's'. Take density of water as ρ_w. Take angle of contact as zero.

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PFM167

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A thin uniform tube is bent into a circle of radius r in the vertical plane. Equal volumes of two immiscible liquids, whose densities are ρ_1 and $\rho_2(\rho_1 > \rho_2)$, fill half the circle. The angle θ between the radius vector passing through the common interface and the vertical is: **[JEE (Main) 2018]**

(1) $\theta = tan^{-1}\pi\left(\frac{\rho_1}{\rho_2}\right)$ (2) $\theta = tan^{-1}\frac{\pi}{2}\left(\frac{\rho_1}{\rho_2}\right)$ (3) $\theta = tan^{-1}\left[\frac{\pi}{2}\left(\frac{\rho_1-\rho_2}{\rho_1+\rho_2}\right)\right]$ (4) None of these

PFM033

2. A cylindrical plastic bottle of negligible mass is filled with 310 *ml* of water and left floating in a pond with still water. If pressed downward slightly and released, it starts performing simple harmonic motion at angular frequency ω. If the radius of the bottle is 2.5 *cm* then ω close to :

(Density of water = 10^3 *kg / m*^{3}) **[JEE (Main) 2019]**

(1) $2.5\sqrt{10}$ *rad s*$^{-1}$ (2) 1.25 *rad s*$^{-1}$ (3) 3.75 *rad s*$^{-1}$ (4) 2.50 *rad s*$^{-1}$

PFM034

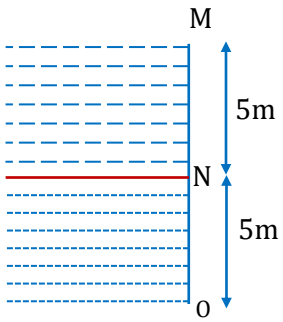
3. An ideal fluid flows (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 *cm* and 4.8 *cm*, respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is: **[JEE (Main) 2020]**

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{3}{4}$ (3) $\frac{81}{256}$ (4) $\frac{9}{16}$

PFM036

4. Two liquids of densities ρ_1 an ρ_2 ($\rho_2 = 2\rho_1$) are filled up behind a square wall of side 10 *m* as shown in figure. Each liquid has a height of 5 *m*. The ratio of the forces due to these liquids exerted on upper part *MN* to that at the lower part *NO* is (Assume that the liquids are not mixing)

[JEE (Main) 2020]



(1) 1/4 (2) 2/3 (3) 1/3 (4) 1/2

PFM037

5. Two narrow bores of diameter 5.0 *mm* and 8.0 *mm* are joined together to form a *U*-shaped tube open at both ends. If this *U*-tube contains water, what is the difference in the level of two limbs of the tube.

[Take surface tension of water $T = 7.3 \times 10^{-2}$ *Nm*$^{-1}$, angle of contact = 0, $g = 10$ *ms*$^{-2}$ and density of water = 1.0×10^3 *kg m*$^{-3}$] **[JEE (Main) 2020]**

(1) 3.62 *mm* (2) 2.19 *mm* (3) 5.34 *mm* (4) 4.97 *mm*

PFM038

For More Join: @IITJEE_Advanced

6. A soap bubble of radius 3 cm is formed inside the another soap bubble of radius 6 cm. The radius of an equivalent soap bubble which has the same excess pressure as inside the smaller bubble with respect to the atmospheric pressure is ..... cm. **[JEE (Main) 2021]**

PFM039

7. Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01 cm^3 of oleic acid per cm^3 of the solution. Then you make a thin film of this solution (monomolecular thickness) of area 4 cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3}$ cm. Then the thickness of oleic acid layer will be $x \times 10^{-14}$ m. Where x is_____.

[JEE (Main) 2021]

PFM041

8. A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be:

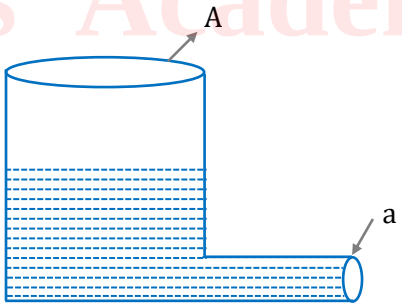
[JEE (Main) 2021]

(1) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$ (2) $\frac{2T}{rJ}$ (3) $\frac{3T}{rJ}$ (4) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

PFM042

9. A light cylindrical vessel is kept on a horizontal surface. Area of base is A. A hole of cross-sectional area 'a' is made just at its bottom side. The minimum coefficient of friction necessary to prevent sliding the vessel due to the impact force of the emerging liquid is ($a << A$) : **[JEE (Main) 2021]**

Rankers Academy JEE



(1) $\frac{A}{2a}$ (2) None of these (3) $\frac{2a}{A}$ (4) $\frac{a}{A}$

PFM044

10. The velocity of upper layer of water in a river is 36 kmh^{-1}. Shearing stress between horizontal layers of water is 10^{-3} Nm^{-2}. Depth of the river is _m.
(Co-efficiency of viscosity of water is 10^{-2} $Pa.s$) **[JEE (Main) 2022]**

PFM045

11. When a ball is dropped into a lake from a height 4.9 m above the water level, it hits the water with a velocity v and then sinks to the bottom with the constant velocity v. It reaches the bottom of the lake 4.0 s after it is dropped. The approximate depth of the lake is : **[JEE (Main) 2022]**

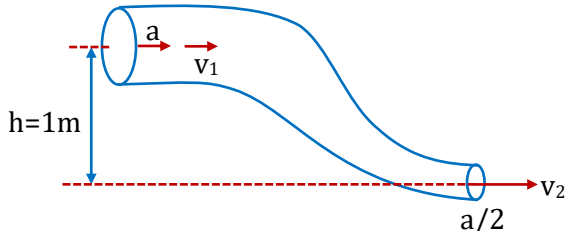
(1) 19.6 m (2) 29.4 m (3) 39.2 m (4) 73.5 m

PFM046

For More Join: @IITJEE_Advanced

12. An ideal fluid of density 800 kgm^{-3}, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6} ms^{-1}$ for x = ...

(Given g = 10 m^{-2})



[JEE (Main) 2022]

PFM047

13. A fully loaded boeing aircraft has a mass of 5.4 × 10^5 kg. Its total wing area is 500 m^2. It is in level flight with a speed of 1080 km/h. If the density of air ρ is 1.2 kg m^{-3}, the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be (g = 10 m/s^2)–

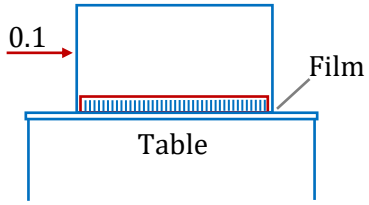
[JEE (Main) 2023]

(1) 16 (2) 6 (3) 8 (4) 10

PFM048

14. A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is 5.0 × 10^{-3} Pl, the speed of block is _____× 10^{-3} m/s.

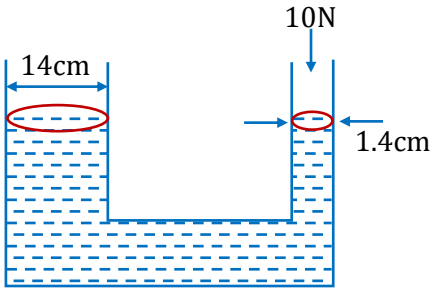
[JEE (Main) 2023]



PFM049

15. A hydraulic press containing water has two arms with diameters as mentioned in the figure. A force of 10 N is applied on the surface of water in the thinner arm. The force required to be applied on the surface of water in the thicker arm to maintain equilibrium of water is _____ N.

[JEE (Main) 2024]



PFM188

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16. A sphere of relative density σ and diameter D has concentric cavity of diameter d. The ratio of $\frac{D}{d}$, if it just floats on water in a tank is:

[JEE (Main) 2024]

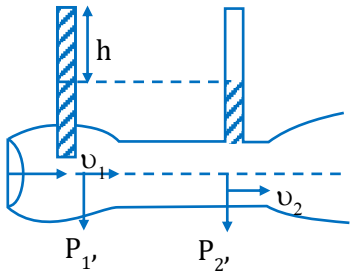
(1) $\left(\frac{\sigma}{\sigma-1}\right)^{\frac{1}{3}}$ (2) $\left(\frac{\sigma+1}{\sigma-1}\right)^{\frac{1}{3}}$ (3) $\left(\frac{\sigma-1}{\sigma}\right)^{\frac{1}{3}}$ (4) $\frac{A}{2a}$

PFM189

17. Given below are two statements :

Statement I : When speed of liquid is zero everywhere, pressure difference at any two points depends on equation $P_1 - P_2 = \rho g\ (h_2 - h_1)$

Statement II : In ventury tube shown $2gh = \upsilon_1^2 - \upsilon_2^2$



In the light of the above statements, choose the most appropriate answer from the options given below.

[JEE (Main) 2024]

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(1) Both Statement I and Statement II are correct.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are incorrect.
(4) Statement I is correct but Statement II is incorrect.

PFM190

18. Two soap bubbles of radius 2 cm and 4 cm, respectively, are in contact with each other. The radius of curvature of the common surface, in cm, is _____ .

[JEE (Main) 2025]

PFM233

19. An air bubble of radius 0.1 cm lies at a depth of 20 cm below the free surface of a liquid of density 1000 kg/m^3. If the pressure inside the bubble is 2100 N/m^2 greater than the atmospheric pressure, then the surface tension of the liquid in SI unit is (use g = 10 m/s^2)

[JEE (Main) 2025]

(1) 0.02 (2) 0.1 (3) 0.25 (4) 0.05

PFM234

20. Water flows in a horizontal pipe whose one end is closed with a valve. The reading of the pressure gauge attached to the pipe is P_1. The reading of the pressure gauge falls to P_2 when the valve is opened. The speed of water flowing in the pipe is proportional to

[JEE (Main) 2025]

(1) $\sqrt{P_1 - P_2}$ (2) $(P_1 - P_2)^2$ (3) $(P_1 - P_2)^4$ (4) $P_1 - P_2$

PFM235

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EXERCISE - JEE (Advanced) PYQ

1. Consider two solid spheres P and Q each of density 8 $gm\ cm^{-3}$ and diameters 1 cm and 0.5 cm, respectively. Sphere P is dropped into a liquid of density 0.8 $gm\ cm^{-3}$ and viscosity $\eta = 3$ poiseulles. Sphere Q is dropped into a liquid of density 1.6 $gm\ cm^{-3}$ and viscosity $\eta = 2$ poiseulles. The ratio of the terminal velocities of P and Q is. **[JEE (Advanced) 2016]**

PFM073

2. A drop of liquid of radius $R = 10^{-2}\ m$ having surface tension $S = \frac{0.1}{4\pi} Nm^{-1}$ divides itself into K identical drops. In this process the total change in the surface energy $\Delta U = 10^{-3}\ J$. If $K = 10^{\alpha}$ then the value of α is **[JEE (Advanced) 2017]**

PFM074

3. Consider a thin square plate floating on a viscous liquid in a large tank. The height h of the liquid in the tank is much less than the width of the tank. The floating plate is pulled horizontally with a constant velocity u_0. Which of the following statements is (are) true ? **[JEE (Advanced) 2018]**

(A) The resistive force of liquid on the plate is inversely proportional to h

(B) The resistive force of liquid on the plate is independent of the area of the plate

(C) The tangential (shear) stress on the floor of the tank increases with u_0.

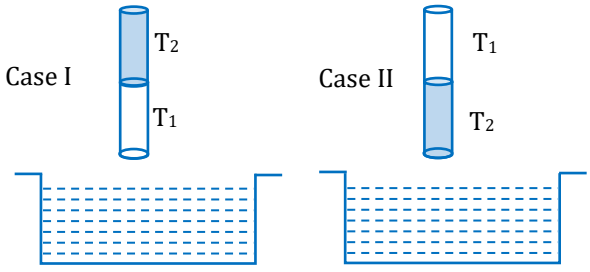
(D) The tangential (shear) stress on the plate varies linearly with the viscosity h of the liquid.

PFM075

4. A cylindrical capillary tube of 0.2 mm radius is made by joining two capillaries T_1 and T_2 of different materials having water contact angles of 0° and 60°, respectively. The capillary tube is dipped vertically in water in two different configurations, case I and II as shown in figure. Which of the following option(s) is(are) correct?

(Surface tension of water = 0.075 N/m, density of water = 1000 kg/m^3, take $g = 10\ m/s^2$)

[JEE (Advanced) 2019]



(A) The correction in the height of water column raised in the tube, due to weight of water contained in the meniscus, will be different for both cases.

(B) For case I, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be more than 8.75 cm. (Neglect the weight of the water in the meniscus)

(C) For case I, if the joint is kept at 8 cm above the water surface, the height of water column in the tube will be 7.5 cm. (Neglect the weight of the water in the meniscus)

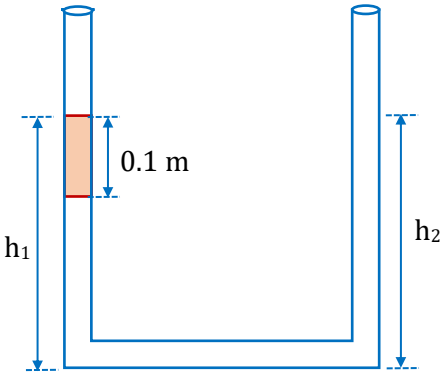
(D) For case II, if the capillary joint is 5 cm above the water surface, the height of water column raised in the tube will be 3.75 cm. (Neglect the weight of the water in the meniscus)

PFM076

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5. An open-ended *U*-tube of uniform cross-sectional area contains water (density $10^3 kg\ m^{-3}$). Initially the water level stands at 0.29 *m* from the bottom in each arm. Kerosene oil (a water-immiscible liquid) of density 800 $kg\ m^{-3}$ is added to the left arm until its length is 0.1 *m*, as shown in the schematic figure below. The ratio $\left(\frac{h_1}{h_2}\right)$ of the heights of the liquid in the two arms is-

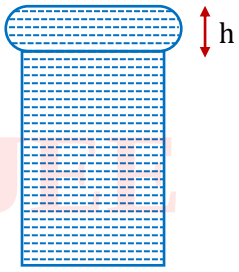
[JEE (Advanced) 2020]



(A) $\frac{15}{14}$ (B) $\frac{35}{33}$ (C) $\frac{7}{6}$ (D) $\frac{5}{4}$

PFM077

6. When water is filled carefully in a glass, one can fill it to a height *h* above the rim of the glass due to the surface tension of water. To calculate *h* just before water starts flowing, model the shape of the water above the rim as a disc of thickness *h* having semicircular edges, as shown schematically in the figure. When the pressure of water at the bottom of this disc exceeds what can be withstood due to the surface tension, the water surface breaks near the rim and water starts flowing from there. If the density of water, its surface tension and the acceleration due to gravity are $10^3 kg\ m^{-3}$, 0.07 Nm^{-1} and 10 ms^{-2}, respectively, the value of *h* (in *mm*) is _____.



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[JEE (Advanced) 2020]

PFM078

7. A train with cross-sectional area S_t is moving with speed v_t inside a long tunnel of cross-sectional area S_0 ($S_0 = 4S_t$). Assume that almost all the air (density ρ) in front of the train flows back between its sides and the walls of the tunnel. Also, the air flow with respect to the train is steady and laminar. Take the ambient pressure and that inside the train to be p_0. If the pressure in the region between the sides of the train and the tunnel walls is *p*, then $p_0 - p = \frac{7}{2N}\rho v_t^2$. The value of *N* is _____.

[JEE (Advanced) 2020]

PFM079

8. A hot air balloon is carrying some passengers, and a few sandbags of mass 1 *kg* each so that its total mass is 480 *kg*. Its effective volume giving the balloon its buoyancy is *V*. The balloon is floating at an equilibrium height of 100 *m*. When *N* number of sandbags are thrown out, the balloon rises to a new equilibrium height close to 150 *m* with its volume *V* remaining unchanged. If the variation of the density of air with height *h* from the ground is $\rho(h) = \rho_0 e^{-\frac{h}{h_0}}$, where $\rho_0 = 1.25\ kg\ m^{-3}$ and $h_0 = 6000\ m$, the value of *N* is _____.

[JEE (Advanced) 2020]

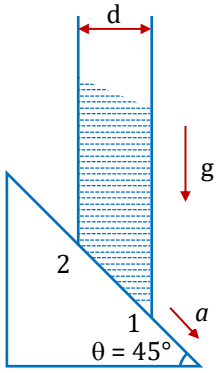
PFM080

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9. A cubical solid aluminium (bulk modulus $=-V\frac{dP}{dV}=70\ GPa$) block has an edge length of 1 m on the surface of the earth. It is kept on the floor of a 5 km deep ocean. Taking the average density of water and the acceleration due to gravity to be $10^3\ kg\ m^{-3}$ and 10 ms^{-2}, respectively, the change in the edge length of the block in mm is ____. **[JEE (Advanced) 2020]**

PFM081

10. A cylindrical tube, with its base as shown in the figure, is filled with water. It is moving down with a constant acceleration a along a fixed inclined plane with angle $\theta = 45°$. P_1 and P_2 are pressures at points 1 and 2, respectively, located at the base of the tube. Let $\beta = (P_1 - P_2)/(\rho g d)$, where ρ is density of water, d is the inner diameter of the tube and g is the acceleration due to gravity. Which of the following statement(s) is(are) correct?



[JEE (Advanced) 2021]

(A) $\beta = 0$ when $a = g/\sqrt{2}$

(B) $\beta > 0$ when $a = g/\sqrt{2}$

(C) $\beta = \frac{\sqrt{2}-1}{\sqrt{2}}$ when $a = g/2$

(D) $\beta = \frac{1}{\sqrt{2}}$ when $a = g/2$

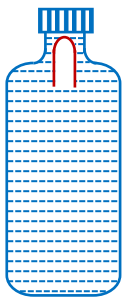
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PFM082

QUESTION STEM FOR QUESTION NO. 11 AND 12

A soft plastic bottle, filled with water of density 1 gm/cc, carries an inverted glass test-tube with some air (ideal gas) trapped as shown in the figure. The test-tube has a mass of 5 gm, and it is made of a thick glass of density 2.5 gm/cc. Initially the bottle is sealed at atmospheric pressure $p_0 = 10^5$ Pa so that the volume of the trapped air is $v_0 = 3.3$ cc. When the bottle is squeezed from outside at constant temperature, the pressure inside rises and the volume of the trapped air reduces. It is found that the test tube begins to sink at pressure $p_0 + \Delta p$ without changing its orientation. At this pressure, the volume of the trapped air is $v_0 - \Delta v$.

Let $\Delta v = X\ cc$ and $\Delta p = Y \times 10^3\ Pa$. **[JEE (Advanced) 2021]**



11. The value of X is ____.

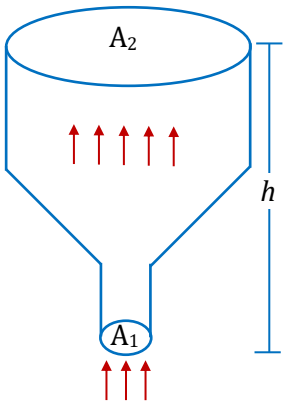
PFM083

12. The value of Y is ____.

PFM084

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13. An ideal gas of density $\rho = 0.2\ kg\ m^{-3}$ enters a chimney of height h at the rate of $\alpha = 0.8 kgs^{-1}$ from its lower end, and escapes through the upper end as shown in the figure. The cross-sectional area of the lower end is $A_1 = 0.1\ m^2$ and the upper end is $A_2 = 0.4\ m^2$. The pressure and the temperature of the gas at the lower end are 600 *Pa* and 300 *K*, respectively, while its temperature at the upper end is 150 *K*. The chimney is heat insulated so that the gas undergoes adiabatic expansion. Take $g = 10\ ms^{-2}$ and the ratio of specific heats of the gas $\gamma = 2$. Ignore atmospheric pressure.



Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2022]**

(A) The pressure of the gas at the upper end of the chimney is 300 *Pa*.

(B) The velocity of the gas at the lower end of the chimney is 40 ms^{-1} and at the upper end is 20 ms^{-1}.

(C) The height of the chimney is 590 *m*.

(D) The density of the gas at the upper end is 0.05 $kg\ m^{-3}$.

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PFM085

14. A bubble has surface tension *S*. The ideal gas inside the bubble has ratio of specific heats $\gamma = \frac{5}{3}$. The bubble is exposed to the atmosphere, and it always retains its spherical shape. When the atmospheric pressure is P_{a1}, the radius of the bubble is found to be r_1 and the temperature of the enclosed gas is T_1. When the atmospheric pressure is P_{a2}, the radius of the bubble and the temperature of the enclosed gas are r_2 and T_2, respectively.

Which of the following statement(s) is(are) correct? **[JEE (Advanced) 2022]**

(A) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{r_1}{r_2}\right)^5 = \frac{P_{a2}+\frac{2S}{r_2}}{P_{a1}+\frac{2S}{r_1}}$.

(B) If the surface of the bubble is a perfect heat insulator, then the total internal energy of the bubble including its surface energy does not change with the external atmospheric pressure.

(C) If the surface of the bubble is a perfect heat conductor and the change in atmospheric temperature is negligible, then $\left(\frac{r_1}{r_2}\right)^3 = \frac{P_{a2}+\frac{4S}{r_2}}{P_{a1}+\frac{4S}{r_1}}$.

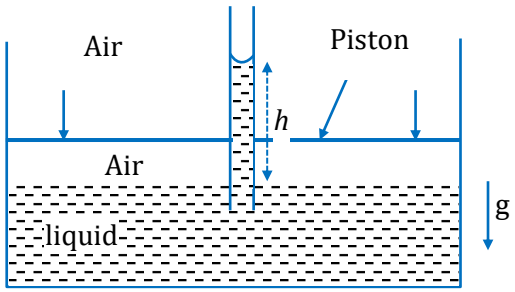
(D) If the surface of the bubble is a perfect heat insulator, then $\left(\frac{T_2}{T_1}\right)^{\frac{5}{2}} = \frac{P_{a2}+\frac{4S}{r_2}}{P_{a1}+\frac{4S}{r_1}}$.

PFM086

For More Join: @IITJEE_Advanced

15. An incompressible liquid is kept in a container having a weightless piston with a hole. A capillary tube of inner radius 0.1 mm is dipped vertically into the liquid through the airtight piston hole, as shown in the figure. The air in the container is isothermally compressed from its original volume V_0 to $\frac{100}{101}V_0$ with the movable piston. Considering air as an ideal gas, the height (h) of the liquid column in the capillary above the liquid level in cm is_____.

[Given: Surface tension of the liquid is 0.075 Nm^{-1}, atmospheric pressure is 10^5 Nm^{-2}, acceleration due to gravity (g) is 10 ms^{-2}, density of the liquid is 10^3 $kg\ m^{-3}$ and contact angle of capillary surface with the liquid is zero] **[JEE (Advanced) 2023]**



PFM176

16. Two large, identical water tanks, 1 and 2, kept on the top of a building of height H, are filled with water up to height h in each tank. Both the tanks contain an identical hole of small radius on their sides, close to their bottom. A pipe of the same internal radius as that of the hole is connected to tank 2, and the pipe ends at the ground level. When the water flows from the tanks 1 and 2 through the holes, the times taken to empty the tanks are t_1 and t_2, respectively. If $H = \left(\frac{16}{9}\right)h$ then the ratio t_1/t_2 is ____.

[JEE (Advanced) 2024]

PFM191

17. spherical soap bubble inside an air chamber at pressure $P_0 = 10^5$ Pa has a certain radius so that the excess pressure inside the bubble is $\Delta P = 144$ Pa. Now, the chamber pressure is reduced to $8P_0/27$ so that the bubble radius and its excess pressure change. In this process, all the temperatures remain unchanged. Assume air to be an ideal gas and the excess pressure ΔP in both the cases to be much smaller than the chamber pressure. The new excess pressure ΔP in Pa is _____.

[JEE (Advanced) 2024]

PFM192

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	D	D	C	D	C	A	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	C	A	D	B	C	C	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	D	A	B	D	B	C	B	A
Que.	31	32								
Ans.	B	A								

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AD	AC	BC	AC	AD	AC	C	A	B	C
Que.	11	12	13							
Ans.	BC	AD	D							

EXERCISE (S)

1. 37.5 N **2.** nH **3.** 40.33 kPa

4. $P_0+3\rho gh$ **5.** $\frac{3l}{2}$ and $\frac{l}{2}$ **6.** $P_0 - \rho a\ell$

7. $H = \frac{L^2\omega^2}{2g}$ **8.** $\frac{5d}{4}$ **9.** 45°

10. 5 **11.** 2 **12.** $+ 29625\ J/m^3, - 30000\ J/m^3$

13. 11 **14.** 1.3 **15.** (i) $\frac{2}{10}\ m/s^2$ (ii) $\sqrt{\frac{m_0 g}{2\rho A}}$

16. $5s$ **17.** 5 **18.** $\frac{dQ}{dt} \propto r^5$

19. 2.52 cm **20.** $h = \frac{mg+4sa}{\rho_w a^2 g}$

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	1	2	2	25	4	3	$100m$
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	363	4	25	1000N	1	4	4	4	1

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9
Ans.	3	6	ACD	ACD	B	3.74	9	4	0.23 to 0.24
Que.	10	11	12	13	14	15	16	17	
Ans.	AC	0.30	10.00	B	CD	25	3	96	

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Important Notes

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